

DUNMAN SECONDARY SCHOOL

CANDIDATE
NAME

CLASS

INDEX
NUMBER

PRELIMINARY EXAMINATION 2025
SECONDARY 4 EXPRESS/ 5 NORMAL ACADEMIC

MATHEMATICS

4052/02

Paper 2

2 hours 15 minutes

SOLUTIONS

Qn	Answer
1(a)	$2x - 1 < \frac{5 + 9x}{2}$ $4x - 2 < 5 + 9x$ $-7 < 5x$ $x > -\frac{7}{5} \text{ or } x > -1.4$
1(b)(i)	$b = \frac{2(n - 5p^2)}{n - p^2}$ $b = 12\frac{2}{3}$
1(b)(ii)	$b = \frac{2(n - 5p^2)}{n - p^2}$ $b(n - p^2) = 2(n - 5p^2)$ $bn - bp^2 = 2n - 10p^2$ $10p^2 - bp^2 = 2n - bn$ $p^2(10 - b) = 2n - bn$ $p^2 = \frac{2n - bn}{10 - b}$ $p = \pm \sqrt{\frac{2n - bn}{10 - b}}$

1(c)

$$\frac{3x}{2x-1} - \frac{2}{5-x} = 7$$

$$\frac{3x(5-x) - 2(2x-1)}{(2x-1)(5-x)} = 7$$

$$15x - 3x^2 - 4x + 2 = 7(10x - 2x^2 - 5 + x)$$

$$-3x^2 + 11x + 2 - 77x + 14x^2 + 35 = 0$$

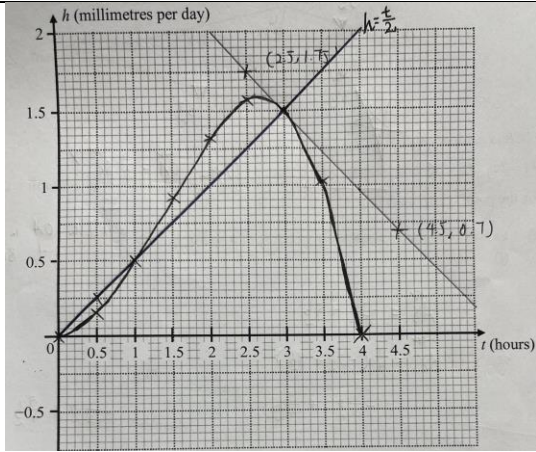
$$11x^2 - 66x + 37 = 0$$

$$x = \frac{-(-66) \pm \sqrt{(-66)^2 - 4(11)(37)}}{2(11)}$$

$$= \frac{66 \pm \sqrt{2728}}{22}$$

$$= 5.37 \text{ (3 s.f.) or } 0.626 \text{ (3 s.f.)}$$

Qn	Answer
2(a)(i)	$\text{Difference} = (3.553 - 3.3064) \text{ million tonnes}$ $= 0.2466 \times 10^6$ $= 2.47 \times 10^5 \text{ tonnes}$
2(a)(ii)	$\text{percentage increase} = \frac{3.553 - 3.040}{3.040} \times 100\%$ $= 16.875\%$ $= 16.9\% \text{ (3 s.f.)}$
2(a)(iii)	$\text{amount of waste in 2022} = 3.553 \times 10^6 \div 84.8 \times 100$ $= 4.19 \times 10^6 \text{ (3 s.f.)}$
2(b)(i)	$\text{deposit} = 20\% \times 120000 = 24000$ $\text{total monthly payment} = 84 \times 1400 = 117600$ $\text{total amount} = 24000 + 117600 = 141600$
2(b)(ii)	$\text{value of car 5 years later}$ $= 120000 \times 0.85^5 = 53244.6375$ $\text{percentage decrease} = \frac{120000 - 53244.6375}{120000} \times 100\%$ $= 55.6\% \text{ (3 s.f.)}$

Question	Answer
3(a)	$h = 0.15$
3(b)	
3(c)	3.5 hours
3(d)(i)	Gradient = $-0.5 (\pm 0.1)$
3(d)(ii)	Since gradient is negative, the daily growth rate decreases with additional sunlight.
3(e)(i)	Straight line passing through (0, 0) and (4, 2)
3(e)(ii)	$\frac{1}{6}t^2(4-t) = \frac{t}{2}$ $4t^2 - t^3 = 3t$ $t^3 - 4t^2 + 3t = 0$

Question	Answer
4(a)(i)	$\overrightarrow{AB} = -2\mathbf{a} + 3\mathbf{b}$ $\overrightarrow{AX} = \frac{1}{4}(-2\mathbf{a} + 3\mathbf{b})$ $= -\frac{1}{2}\mathbf{a} + \frac{3}{4}\mathbf{b}$
4(a)(ii)	$\overrightarrow{OX} = 2\mathbf{a} - \frac{1}{2}\mathbf{a} + \frac{3}{4}\mathbf{b}$ $= \frac{3}{2}\mathbf{a} + \frac{3}{4}\mathbf{b}$
4(b)	$\overrightarrow{BY} = 3(2\mathbf{a}) = 6\mathbf{a}$ $\overrightarrow{XY} = \overrightarrow{XB} + \overrightarrow{BY}$ $= \frac{3}{4}(-2\mathbf{a} + 3\mathbf{b}) + 6\mathbf{a}$ $= \frac{9}{2}\mathbf{a} + \frac{9}{4}\mathbf{b}$
4(c)	$\overrightarrow{OX} = \frac{1}{3}\overrightarrow{XY}$ Since OX is parallel to XY, and X is a common point, O, X and Y lie on a straight line.
4(d)	$\frac{\text{Area of triangle } OBX}{\text{Area of triangle } OAB} = \frac{BX}{AB} = \frac{3}{4}$ $\frac{\text{Area of triangle } OAB}{\text{Area of triangle } ABC} = \frac{OA}{BC} = \frac{2}{3} = \frac{4}{6}$ $\frac{\text{Area of triangle } OBX}{\text{Area of triangle } OABC} = \frac{3}{4+6} = \frac{3}{10}$

4(e)

$$\overrightarrow{OC} = 3\mathbf{a} + 3\mathbf{b}$$

$$\overrightarrow{OW} = \frac{2}{5}(3\mathbf{a} + 3\mathbf{b})$$

$$= \frac{6}{5}\mathbf{a} + \frac{6}{5}\mathbf{b}$$

Question	Answer
5(a)	$3^2 = 8^2 + 7^2 - 2(8)(7)\cos \angle EFI$ $\cos \angle EFI = \frac{8^2 + 7^2 - 3^2}{2(8)(7)}$ $= \frac{13}{14}$ $\angle EFI = \cos^{-1}\left(\frac{13}{14}\right) = 21.7867\dots = 21.8^\circ$
5(b)	$\text{vol of prism} = \left(8 \times 3 + \frac{1}{2} \times 7 \times 8 \times \sin 21.7867^\circ\right) \times 12$ $= 412.70717\dots$ $= 412.71 \text{ m}^3$
5(c)	$\text{height} = 3 + 7 \sin 21.8^\circ$ $= 5.5995\dots$ $= 5.60 \text{ m (3 s.f.)}$
5(d)	Let X be the point vertically below J. $CX = 7 \cos 21.8^\circ = 6.4994\dots$ $BX = \sqrt{12^2 + 6.4994^2} = 13.647\dots$ $\tan \angle JBX = \frac{5.5995}{13.647}$ $\angle JBX = \tan^{-1}\left(\frac{5.5995}{13.647}\right)$ $= 22.3^\circ \text{ (1 d.p.)}$

Question	Answer
6(a)	$\angle OCT = 90^\circ$ (tangent $\perp$ radius) $\angle AOC = 360^\circ - 90^\circ - 90^\circ - 32^\circ$ ($\angle$ sum of quadrilateral) $= 148^\circ$ $\angle ADC = 148^\circ \div 2$ ($\angle$ at centre = $2\angle$ at circumference) $= 74^\circ$
6(b)	$\angle CBA = 180^\circ - 74^\circ$ ($\angle$ s in opposite segment) $= 106^\circ$ $\angle OCA = \frac{180^\circ - 148^\circ}{2}$ (base $\angle$ s of isosceles triangle) $= 16^\circ$ $\angle OCB = 16^\circ + 38^\circ$ (base $\angle$ s of isosceles triangle) $= 54^\circ$ $\angle OCB + \angle CBA = 106^\circ + 54^\circ$ $= 160^\circ \neq 180^\circ$ Therefore, by converse of interior angles, OC is not parallel to AB .
6(c)	$\angle AOB = 38^\circ \times 2$ ($\angle$ at centre = $2\angle$ at circumference) $= 76^\circ$ Area of sector $= \frac{76}{360} \times \pi(10^2) = \frac{190}{9} \pi$ Area of triangle $= \frac{1}{2}(10)(10)\sin 76^\circ = 48.5147\dots$ Area of shaded region $= \frac{190}{9} \pi - 48.5147 = 17.8 \text{ cm}^2$ (3 s.f.)

Question	Answer
7(a)(i)	$n + 20$
7(a)(ii)	Product of top left and bottom right = $n(n + 20) = n^2 + 20n$ Product of top right and bottom left = $(n + 2)(n + 18) = n^2 + 20n + 36$ Difference = $(n^2 + 20n + 36) - (n^2 + 20n) = 36$
7(a)(iii)	Sum = $n + (n + 2) + (n + 10) + (n + 18) + (n + 20) = 5n + 50$ Let $5n + 50 = 1715 \Rightarrow n = 333$ If $n = 333$, the cross will be $\begin{array}{ccc} 333 & \dots & 335 \\ & \dots & 343 \\ 351 & \dots & 353 \end{array}$ However, $333 = 9 \times 37$, the first number will be in the last column of the number grid, therefore, the sum cannot be 1715.
7(b)(i)	$T_{10} = 2(10)^2 - 10 + 3 = 193$
7(b)(ii)	$\begin{aligned} T_{n+1} &= 2(n+1)^2 - (n+1) + 3 \\ &= 2(n^2 + 2n + 1) - n - 1 + 3 \\ &= 2n^2 + 3n + 4 \\ D &= 2n^2 + 3n + 4 - (2n^2 - n + 3) \\ &= 4n + 1 \end{aligned}$
7(b)(iii)	Since D is a linear expression in n and the coefficient of n is 4, it means that the difference increases by 4 each time n increases by 1.

Question	Answer
8(a)(i)	7.5 hours (± 0.1)
8(a)(ii)	Q1 = 6, Q3 = 9.4 IQR = $9.4 - 6 = 3.4$ (± 0.1)
8(b)	32 students spent 5 hours or less. 168 students spent at least 5 hours. Probability = $\frac{168}{200} = \frac{21}{25} = 0.84$
8(c)	In general, teenagers in Country X spent less daily screen time with median of 6.5 hours compared to Singapore's median of 7.5 hours. The amount of daily screen time spent by teenagers in Country X has a larger spread with an IQR of 5.2 hours as compared to that of Singapore with IQR of 3.4 hours.

Question	Answer
9(a)	Arc length = $2\pi(2) \times \frac{1}{5} = \frac{4}{5}\pi = 2.513... = 2.5$ m (1 d.p.)
9(b)	Volume of air = $\pi(2.5)^2 \times 0.5 \times 50 \div 60 = 8.1812... = 8.18$ (3 s.f.)
9(c)	<u>For air circulation</u> Minimum airflow requirement = $30 \times 20 \times 8 \div 5 = 960$ m ³ / min VFR = $\pi(3)^2 \times 0.6 = 5.4\pi = 16.9646...$ Total airflow per minute of fans = $VFR \times 70 \times 2 = 756\pi = 2375.044...$ Therefore, $2375.044... > 960$, minimum airflow requirement is met. <u>For comfortable cooling environment</u> VPS = $5.4\pi \times 45 \div 60 = 4.05\pi$ Air velocity = $4.05\pi \div \pi(3)^2 = 0.45$ m/s > 0.3 m/s Therefore, comfortable cooling environment is met. Hence, the recommendation is suitable.